Gwon Ryul Han

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Education

Ulsan National Institute of Science and Technology (UNIST) Ulsan, *Republic of Korea*
B.S. in Physics *March 2024 – Present*

Research Interests

- Quantum Information Theory, Quantum Metrology, Quantum State over Spacetime
- Mixed State Topological Order, Anyon Theory

Publications

[1] Gwon Ryul Han, Hyukgun Kwon, and Seok Hyung Lie. *Quantum Metrology for Signals with an Unknown Causal Structure*. 2026. arXiv: 2609.28066 [quant-ph]. URL: https://arxiv.org/abs/2609.28066.

Research Experience

Quantum Information Theory Group

Undergraduate intern Researcher (Advised by Seok Hyung Lie) UNIST *June 2024 – Present*

- **Research Topic**: Quantum Metrology, Quantum State over Spacetime

Topological Matter Theory Group

Undergraduate intern Researcher (Advised by Gilyoung Cho) KAIST *June 2026 – Present*

- **Research Topic**: Mixed State Topological Order, Anyon Theory (related to Quantum Error Correction Code)